

Final Presentation and Notebook Submission

Canvas submission due before presentation day. Present live from your notebook.

You have built four phases of a data science project. This document tells you how to wrap it up, what to submit, and how you will be graded.

There are two deliverables. Both go on Canvas. Both matter.

- A final consolidated Jupyter Notebook that tells the clean story of your project from start to finish
- A slide deck that serves as the visual anchor for your live presentation

On presentation day you will present live from your notebook. Your slides are not a script. They are a visual guide for the class. You explain. The notebook proves it.

Logistics

Each team has 4 minutes to present followed by 1 minute for one question from the class. That is tight. It is supposed to be tight. Constraints force clarity.

*4 minutes presenting. 1 minute for one class question. Hard stop.
Both partners must speak. A presentation where one person presents alone will lose delivery points automatically.*

With 25 teams and 2 hours, there is no slack in the schedule. Show up on time, have your notebook open and ready, and know your slides.

Part 1: The Slide Deck

You have 7 slides. Each slide has one job.

Any format is acceptable. PowerPoint, Google Slides, Keynote, Canva. It does not matter. What matters is that it is readable from the back of the room and submitted as a PDF or a shareable link on Canvas.

Slide by Slide Breakdown

Slide	Title	What Goes On It	Notebook Section
1	Title Slide	Project title, your names, research question in one sentence	README

2	The Data	Dataset source, shape, target variable, one key EDA visualization	Phase 1 + Phase 2
3	EDA Findings	The relationship that motivated your model. Scatter plot with correlation value.	Phase 2
4	Baseline and Model	Constant model RMSE vs SLR RMSE. Your model equation written out.	Phase 3 Part 2 + 4
5	Did It Improve?	Unified comparison table. Best model RMSE on test set vs baseline.	Phase 4 Part 5
6	What It Still Gets Wrong	Residual plot from Phase 4. One honest sentence about what the model misses.	Phase 4 Part 4
7	What Comes Next	Two concrete next steps. One additional feature. One modeling technique.	Phase 4 Part 6e

Design Rules

- One main idea per slide. If you are tempted to add a second point, make a second slide or cut it.
- Every plot must have a title and labeled axes. A plot without labels is unreadable to the class and will cost you points.
- No walls of text. If you have more than four lines of text on a slide, you have too much. Cut it.
- Font size 24 or larger for body text. Your classmates in the back row need to read it.
- Consistent visual style throughout. Pick a color scheme and stick with it.

The single most common design failure: copying a table or plot directly from your notebook without cleaning it up. Notebook output is for you. Slides are for an audience. Remake your plots if you need to. Simplify your tables. Your audience has 30 seconds per slide.

Think of your slides as the whiteboard behind you while you talk. They show the plot. You explain what it means. They show the table. You explain what changed and why. The words come from you, not the slide.

Part 2: The Final Consolidated Notebook

This is not your Phase 1 notebook plus your Phase 2 notebook plus your Phase 3 notebook stapled together. That is a mess and it will read like one.

This is a single clean notebook that tells the story of your project from start to finish. It shows only the decisions you actually made, with brief explanations of why you made them. Dead ends, failed experiments, and abandoned features are mentioned in a sentence but not shown in code.

Think of this notebook as the report you would hand to a data science manager at a company. They want to understand what you did and why. They do not want to watch you try things. They want to see the final pipeline and understand the reasoning behind it.

Notebook Structure

Your notebook must follow this structure exactly. Section numbers and titles are required. Use markdown headers.

Section	Title	What to Include	What to Cut
0	README Header	Research question, target, features, final dataset shape, one sentence per phase summarizing what you found	Nothing, this is required
1	Data Loading and Cleaning	Load data, handle missing values, final shape. One markdown cell explaining your dataset and where it came from.	Every failed cleaning attempt. Show the final pipeline only.
2	EDA	Three to five visualizations that tell a story. The scatter plot that motivated your feature. The correlation value. A brief written interpretation of each.	Every exploratory plot you made. Keep only the ones that informed a decision.
3	Baseline Model	Constant model MSE and RMSE. The written interpretation of what that number means in your target variable units.	No cuts. This section is already lean.
4	Simple Linear Regression	Model equation written out. Sklearn verification. Scatter plot with regression line. Phase 3 residual plot. Written interpretation of both.	Manual implementation helper functions can be collapsed or removed. Keep the results, not the scaffolding.
5	Train/Test Split	Split code. Print of train and test shapes. One markdown cell explaining why you are doing this and what it fixes.	No cuts needed.

6	Feature Engineering	For each engineered feature: one sentence justification, the transformation code, before and after evidence. Keep only features that made it into your final model.	Dead ends and features you tried and discarded. Mention them in one sentence but do not show the code.
7	Final Model and Evaluation	Retrained model on engineered features. Unified comparison table. Residual plot on test set. Actual vs predicted plot on test set. Full written interpretation.	Nothing. This is the heart of the notebook.
8	Reflection and Next Steps	Two paragraphs. What your model still gets wrong and why. What you would do next with more time or data.	Generic statements like 'the model performed well'. Be specific.

Writing Standards

Every section must contain at least one markdown cell that explains what you are doing and why. Code without explanation is not a narrative notebook. It is a script.

- Write in plain English. Explain your reasoning as if your reader is a smart person who has not seen your data before.
- Interpret every plot. A plot with no written interpretation will receive zero credit for that section.
- Reference your results by number. Do not write 'the RMSE improved.' Write 'the RMSE dropped from 14.3 to 11.7 on the test set, a 18 percent improvement over the constant model baseline.'
- Be honest about failures. A model that barely beat the baseline is not a failed project. It is a finding. Name it and explain it.

What a Clean Narrative Notebook Looks Like

Here is the structure of a well-written section. Use this as your template for every section.

Example: Feature Engineering Section

Markdown cell: 'Our Phase 3 residual plot showed a clear curve pattern, suggesting the relationship between study_hours and exam_score is not purely linear at higher values. We tested a log transformation of study_hours to see if it linearized the relationship.'

Code cell: transformation code

Output cell: before and after scatter plots

Markdown cell: 'The log transformation reduced the curvature in the residual plot and improved test RMSE from 11.7 to 10.2. We kept it in the final model.'

Notice what is not there. There is no code for the five other transformations you tried. There is no plot for the interaction term that made things worse. Those happened. You mention them in one sentence. You do not show them.

Part 3: How You Will Be Graded

The presentation and notebook are graded together as one combined submission. The notebook is the evidence. The presentation is the argument. Both matter and neither works without the other.

Grade Breakdown

Dimension	What We Are Looking For	Full Credit	Partial / No Credit
Content (60 pts)	Research question is clearly stated. EDA findings are cited with evidence. Model decisions are justified. Comparison table is present and correctly interpreted. Limitations are named honestly.	All five content elements present and accurate	Missing elements, incorrect numbers, or no interpretation
Design (25 pts)	Slides are readable from the back of the room. One main idea per slide. Plots have titles and labeled axes. No walls of text. Consistent visual style throughout.	Clean, readable, one idea per slide, all plots labeled	Cluttered slides, unlabeled plots, inconsistent formatting
Delivery (15 pts)	Both partners speak. Presenters explain from the notebook, not read from slides. Questions are answered with reference to specific results.	Both speak, explain reasoning, handle one question	One partner presents alone, or presenters read slides verbatim
TOTAL	100 points		

Content: The Five Things We Are Looking For

Content is 60 points. Here is exactly what earns those points.

- **Your research question is clearly stated and your target variable is explicitly named.**
 - We should know within the first 60 seconds what you were trying to predict and why it matters.
- **Your EDA findings are cited with evidence, not just asserted.**
 - Do not say 'we found a strong relationship.' Say 'the correlation between study_hours and exam_score was 0.73, which we can see in slide 3.'
- **Your model decisions are justified.**
 - Why did you choose that feature in Phase 3? Why did you engineer that transformation in Phase 4? The answer should always be 'because our EDA showed...' or 'because our Phase 3 residuals showed...'
- **Your comparison table is present and correctly interpreted.**
 - The numbers must match your notebook. We will check.
 - You must interpret the table in plain English using the actual units of your target variable.
- **Your limitations are named honestly.**
 - What does your residual plot still show? What does your model still get wrong?
 - A presentation with no acknowledged limitations suggests you do not understand your own model.

Design: What Readable Means

Design is 25 points. We are not grading you on being a graphic designer. We are grading you on whether your slides communicate clearly to a room full of people.

- One main idea per slide. Your audience has 30 seconds per slide. Give them one thing to understand.
- All plots have titles and labeled axes. This is also a content requirement. An unlabeled plot is an uninterpretable plot.
- No walls of text. If a slide has more than four lines of text, cut it or split it.
- Readable font size. If your classmates in the back row cannot read it, it does not exist.
- Consistent visual style. This does not mean beautiful. It means not jarring.

Delivery: What We Are Listening For

Delivery is 15 points. This is the lowest weight dimension but it is still 15 points and it is entirely within your control.

- Both partners speak. This is a hard requirement. One person presenting alone automatically loses all delivery points.
- You explain from the notebook, not read from the slides. We want to hear your reasoning, not your slide text.
- When a question is asked, you answer it by pointing to a specific result in your notebook. 'I think' is a weak answer. 'Our test RMSE was X which shows...' is a strong answer.

Part 4: Canvas Submission Checklist

Submit both of the following to Canvas before presentation day. Late submissions will not be accepted because we need to load your notebook before your slot.

- Final consolidated Jupyter Notebook (.ipynb) with all outputs run and visible
- Slide deck submitted as PDF or shareable link

Before you submit, go through this checklist:

Notebook: Kernel restarted and all cells run top to bottom without errors

Notebook: Every section has at least one markdown cell explaining what and why

Notebook: Every plot has a title and labeled axes

Notebook: README header is complete with research question, target, features, and final dataset shape

Notebook: Unified comparison table is present with train RMSE, test RMSE, and improvement over baseline

Slides: Seven slides, one main idea each

Slides: All plots have titles and labeled axes

Slides: Both partner names on the title slide

Slides: Submitted as PDF or shareable link, not an editable file

Timing: You have practiced out loud and finished in under 4 minutes